

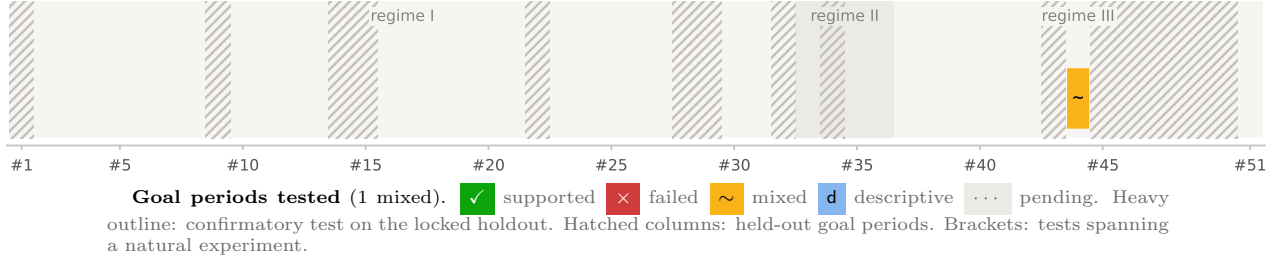
H23: Distilling the village into a fine-tuned leader copies its vocabulary but transforms its plans

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1.5/5 Usefulness 1.5/5

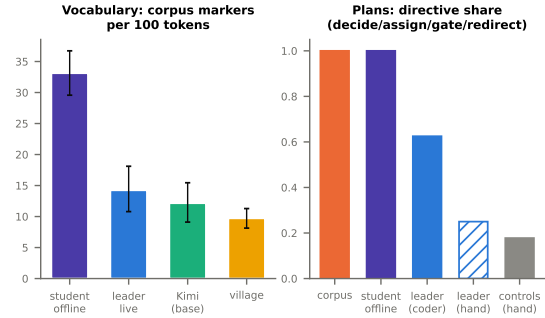
Bottom line. The fine-tuned leader carried a trace of its corpus’s addressing style into the room but none of its plans, which were not transmitted rather than transformed, and its corpus was mostly its own base model talking to itself.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

From the agents’ terminals: the deployed leader (agent 28 in #44, agent 30 in #45) is base Kimi K2.6 LoRA-tuned on 64 rows, 80% of them Kimi’s own replies to 20 synthetic village scenarios under a leadership prompt; no village chat was a target. On its 16 live #44 messages (G44, exploratory, not blind) vs 45 same-window messages: corpus-distinctive vocabulary 14.2 vs 10.0 per 100 tokens ($p = 0.011$), mostly agent addressing and markup; phrase copy n.s.; lexical copy information 0.012 bits per word. Plans: directive share 0.62 (hand-coded 0.25) vs the corpus’s 1.00; excess plan copy -0.04 vs a perfect-copy 0.23, transformation information at the null: *not transmitted*, not transformed. Not closer to the corpus than to its base in embeddings; base-echo and latest-chat rivals stand. #45 test written, **not run**.



Observables. (a) Corpus-distinctive n-grams per 100 tokens (95% CI): the offline student (sibling checkpoints on training-format prompts) sits far above the room; the live leader keeps about a fifth of that excess, near base Kimi in the same window ($n = 6$). (b) Directive share of plan acts: corpus and offline student 100%; the live leader (hatched: post hoc hand coding) sits near the room.

PHYSICAL MODEL

Model 08 (Kolchinsky & Corominas-Murtra), distillation variant: a channel from the corpus X to the leader’s live output Y on a shared alphabet (word n-grams, or 10 keyword-coded plan acts). Live message m is paired with the corpus rows whose scenario is nearest its context, $p_C(x | m)$. d : binary KL; r_x : probability of reproducing x ; a, b : rates at which the leader and untrained same-room speakers emit corpus n-grams; c : copy fraction.

$$I(X; Y) = I_{\text{copy}} + I_{\text{transform}}$$
$$I_{\text{copy}} = \sum_x p(x) d(r_x \| p_Y(x)) \mathbf{1}[r_x > p_Y(x)]$$
$$p(x, y) = \frac{1}{n} \sum_m p_C(x | m) \delta_{y, y_m}, \quad c = \sum_x p(x, x)$$
$$I_{\text{copy}}^{\text{lex}} = d(a \| b) \text{ bits per n-gram}$$

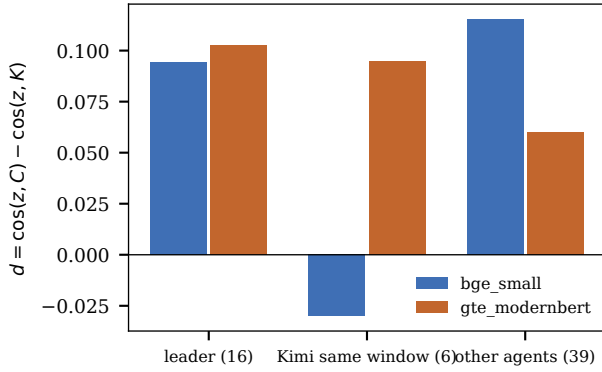
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For anyone distilling a swarm into a leader model, the lesson is mundane: SFT on a few dozen synthetic crisis scenarios carried the surface style (naming agents, markup, terse format) into deployment but not the decision policy, because live situations (routine project work) looked nothing like the training scenarios. That coverage failure is standard ML, not a swarm insight. The decomposition adds one useful distinction (not transmitted vs transformed) and a calibration (the perfect-copy reference), but 16 messages and a coder at $\kappa \approx 0.45$ measure little. The real surprise is the pipeline: the “village distillation” was the base model distilling itself.

H23-leader-distillation-copy · generated 2026-10-04 11:10 by infra/summaries/build_summaries.py · card hypotheses/H23-leader-distillation-copy/README.md · ratings: calibrated by coordinator 2026-10-04 (from H23 agent (suggested))

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b	Verdict (1b)
P1a corpus vocabulary > room	14.2 vs 10.0 /100 tokens ($p = 0.011$); unchanged	yes
P1b bigram copy room	> 0.064 vs 0.055 ($p = 0.17$); unchanged	inconclusive
P2a plans not copied	directive 0.62 vs corpus 1.00; unchanged	yes
P2b plans transformed	$I_{transform}$ within null; unchanged (shared copy_info, 1,022 numbers identical)	no (not transmitted)
P3a closer to corpus than base	bge: leader > Kimi $p = 0.08$ → gte: $p = 0.43$ (Kimi as corpus-like)	no, both models
Checkpoint labels	DQ6 times relabel 0 of 27 leader messages	unchanged
R1 base echo rejected	no → no, and stronger (H46 rival stands style, H32 weakest source)	



Second figure. Embedding-level proximity $d = \cos(z, \bar{z}_{corpus}) - \cos(z, \bar{z}_{Kimi\ field})$ in regime-III whitened space for the leader’s 16 live messages, the same-window Kimi K2.6 (6) and the other #best

agents (39), in bge-small (round 1) and gte-modernbert (round 1b).

GOAL PERIODS IN DETAIL

G44 ~ 1b: mixed (unchanged; copy_info and DQ6 checkpoints i...

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

16 leader messages in one afternoon; 31/64 corpus rows recovered; plan coder $\kappa \approx 0.4-0.5$. The #45 confirmatory script would include agent 30’s first ~11 min under a wrong model string (DQ6): an amendment is proposed, not applied. Native tests (#26, #35) not run.

ROUND 2

Amend `confirm_g45.py` with the DQ6 checkpoint window and both embedding models, then ask Vivian about #45. Run the #26 and #35 leader-copy natives.

- **Essence.** What does fine-tuning transmit, compared with prompting?
- **H23-R1** Style transmits through weights, decisions through context: the leader’s influence came from its role prompt and position, not its fine-tune.

H23-leader-distillation-copy · generated 2026-10-04 11:10 by infra/summaries/build-summaries.py · card hypotheses/H23-leader-distillation-copy/README.md · ratings: calibrated by coordinator 2026-10-04 (from H23 agent (suggested)) (page 2)